

Shiyang Li

✉ galianli0815@gmail.com 📍 Minneapolis, MN, USA 🌐 github.com/NKU-Yang

🌐 [NKU-Yang.github.io](https://github.com/NKU-Yang) 🎓 [Google Scholar](https://scholar.google.com/citations?user=0000-0003-3046-0414) 🆔 0000-0003-3046-0414

Interests: GPU Programming, Computer Architecture, Agentic MLSys, EDA

Skills: C/C++, CUDA, Python, Rust, Linux, Git, Docker, LaTeX, oneAPI/SYCL

🎓 Education

- **University of Minnesota-Twin Cities** SEPT.2025 – Present
Ph.D. in Computer Science
Advisors: Prof. Caiwen Ding & Prof. Pen-Chung Yew
- **Nankai University Master of Engineering** SEPT.2022 – JUNE.2025
Major in Computer Technology
Advisor: Prof. Gong Xiaoli
- **Nankai University Bachelor of Engineering** SEPT.2018 – JUNE.2022
Major in Internet of Things Engineering

📖 Publications

- Xingze Liu, Jialin Dong, Xinyu Liu, Lizhi Wang, **Shiyang Li**, Haoran Li, Xiaoli Gong, Jin Zhang, Pen-Chung Yew, “**PagePilot: Synergizing Heterogeneous Backend Devices with Reusability-Aware Page Offloading**,” *ACM Transactions on Architecture and Code Optimization (TACO)*, 2026. [Paper](#). [doi:10.1145/3845616](https://doi.org/10.1145/3845616). [Code](#).
- **Shiyang Li**, Guangyan Sun, Jinwei Tang, Yanzhi Wang, Mingyi Hong, Caiwen Ding, “**SparseDitto: Customizing GPU Kernels for Different Sparsity Patterns with LLM-Based Agentic System**,” *arXiv preprint arXiv:2608.05033*, Aug. 2026.
- Yuebo Luo, **Shiyang Li**, Yifei Feng, Vishal Kancharla, Shaoyi Huang, Caiwen Ding, “**GSR-GNN: Training Acceleration and Memory-Saving Framework of Deep GNNs on Circuit Graph**,” *In Proceedings of the 63rd ACM/IEEE Design Automation Conference (DAC '26)*, June 2026, San Francisco, USA. [Paper](#).
- **Shiyang Li**, Zijian Zhang, Winson Chen, Yuebo Luo, Mingyi Hong, Caiwen Ding, “**StitchCUDA: An Automated Multi-Agents End-to-End GPU Programming Framework with Rubric-based Agentic Reinforcement Learning**,” *International Conference on Machine Learning (ICML 2026)*. [arXiv:2603.02637](#). [Code](#).
- **Shiyang Li**, Haoyang Chen, Mattia Fazzini, Caiwen Ding, “**CUDABeaver: Benchmarking LLM-Based Automated CUDA Debugging**,” *arXiv preprint arXiv:2605.08455*, May 2026. [Code](#).
- **Shiyang Li**, Zijian Zhang, Guangyan Sun, Yuebo Luo, Winson Chen, Yanzhi Wang, Mingyi Hong, Caiwen Ding, “**CUDAHercules: Benchmarking Hardware-Aware Expert-level CUDA Optimization for LLMs**,” *arXiv preprint arXiv:2605.08467*, May 2026.
- Ben Mechels, Ryan Billmeyer, Alexander Chen, **Shiyang Li**, Caiwen Ding, “**FPTC: A Fast Parallel Transform-based Codec for Efficient Asymmetric Signal Compression**,” *arXiv preprint arXiv:2605.01086*, May 2026.
- Zijian Zhang, Rong Wang, **Shiyang Li**, Yuebo Luo, Mingyi Hong, Caiwen Ding, “**CudaForge: An Agent Framework with Hardware Feedback for CUDA Kernel Optimization**,” *arXiv preprint arXiv:2511.01884*, Oct. 2025.
- Yuebo Luo, **Shiyang Li**, Junran Tao, Kiran Gautam Thorat, Xi Xie, Hongwu Peng, Nuo Xu, Caiwen Ding, Shaoyi Huang, “**DR-CircuitGNN: Training Acceleration of Heterogeneous Circuit Graph Neural Network on GPUs**,” *In Proceedings of the 39th ACM International Conference on Supercomputing (ICS '25)*, June 2025, Salt Lake City, USA. [Paper](#).
- **Shiyang Li**, Ruiqi Tang, Jingyu Zhu, Ziyi Zhao, Xiaoli Gong, Wenwen Wang, Jin Zhang, Pen-Chung Yew, “**Liberator: A Data Reuse Framework for Out-of-Memory Graph Computing on GPUs**,” *IEEE Transactions on Parallel and Distributed Systems (TPDS)*, 34(6):1954–1967, 2023. [Code](#).
- **Shiyang Li**, Jingyu Zhu, Jiaxun Han, Yuting Peng, Zhuoran Wang, Xiaoli Gong, Gang Wang, Jin Zhang, Xuqiang Wang, “**OneGraph: A Cross-Architecture Framework for Large-Scale Graph Computing on GPUs Based on oneAPI**,” *CCF Transactions on High Performance Computing (CCF-THPC)*, 6(2):179–191, 2024.

👨‍🏫 Teaching Experience

- **Teaching Assistant of Computer Architecture (CSCI 4203)** Fall 2025
University of Minnesota-Twin Cities. Responsible for leading discussion sections and grading assignments on CPU pipelining, caching, and memory hierarchy.
- **Teaching Assistant of Principles of Computer Organization** Spring 2023
Responsible for teaching experimental courses; assisted students in gradually learning how to use Verilog to implement a 5-level pipeline CPU.

- › Teaching Assistant of Introduction to Parallel Programming

Responsible for teaching programming courses; taught students how to code with CUDA on NVIDIA GPUs and implemented the Gauss Elimination algorithm with CUDA.

Spring 2022
- › Teaching Assistant of Operating System

Responsible for teaching experimental courses and guiding junior undergraduate students in mastering Linux systems and virtual machines. Supported students in developing a toy OS, with emphasis on threads, processes, memory, and file systems.

Autumn 2020, 2021, 2022

 Work Experience

Alibaba Cloud, Shanghai, China

System Engineer Intern

JUL.2021 – SEPT.2021

Container and OS kernel development with Rust

- › Developed Linux kernel modules with Rust for security containers.
- › Reworked the PCIe device registry process in Hypervisor and mapped device memory through MMIO.
- › Implemented NVIDIA GPU passthrough and hot-plug support in security containers with VirtIO.

 Awards

China National Scholarship (only one in college per year)	DEC.2023
Outstanding Student Leader of Nankai University	SEPT.2023
First-class Scholarship for Graduate Students of Nankai University (top 10%)	OCT.2023
Nankai University Special Scholarship (3 times)	2022–2024